

MID

Course: Data Structures (CS101)

Set: A | Time: 2 Hours | Max Marks: 37 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
attend all the quetsions

SECTION A: MCQs

1. Which layer of the OSI model is responsible for logical addressing (IP addresses)? [2 Marks]

- a) Physical Layer
- b) Data Link Layer
- c) Network Layer
- d) Transport Layer

2. Which of the following protocols is considered connection-oriented? [2 Marks]

- a) UDP
- b) HTTP
- c) TCP
- d) ICMP

3. What is the primary function of a modem in a computer network? [2 Marks]

- a) To route data packets between networks.
- b) To modulate and demodulate signals for transmission over analog lines.
- c) To filter network traffic and improve security.
- d) To connect multiple devices within a local area network.

4. A MAC address uniquely identifies a device on a network segment. What is its typical length? [2 Marks]

- a) 16 bits
- b) 32 bits
- c) 48 bits
- d) 128 bits

5. Which of the following transmission media is least susceptible to electromagnetic interference (EMI)? **[3 Marks]**
- a) Coaxial Cable
 - b) Unshielded Twisted Pair (UTP)
 - c) Fiber Optic Cable
 - d) Shielded Twisted Pair (STP)
6. In the context of computer networks, what does LAN stand for? **[2 Marks]**
- a) Local Area Network
 - b) Large Access Node
 - c) Link Area Node
 - d) Low-latency Access Network

SECTION B: SHORT ANSWERS

7. Define the primary purpose of the Transport Layer in the OSI model. **[3 Marks]**
8. Differentiate between a hub and a switch based on how they handle network traffic. **[5 Marks]**
9. Explain the fundamental concept of 'packet switching' in computer networks. **[3 Marks]**
10. Briefly describe two common network topologies and their main characteristics. **[6 Marks]**
11. Discuss the role of DNS (Domain Name System) in network communication. **[7 Marks]**

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